CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Berlin • EU Blue Card linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building AI agents and LLM products from the ground up, from RAG, embeddings and vector search to full-stack delivery and cloud deployment on Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by end-to-end ownership at an early-stage startup, from first concept to deployed product.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Built the core product of an early-stage startup from the ground up as sole developer: a knowledge-extraction system owned end-to-end, from data pipelines (ingestion, chunking, prompt engineering, indexing) to the deployed system, reporting directly to the CTO.
- Built AI agents in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, evaluated against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Built a scalable, deterministic, code-based pipeline instead of LLM calls for a second, high-volume data stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed a Model Context Protocol (MCP) query service with millisecond-latency retrieval over the knowledge graph, integrated into Claude Code for agentic development workflows.
- Shipped Expert Interface, a full-stack graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LLM APIs (LiteLLM), RAG, Vector Search, GraphRAG, MCP, FastAPI, Postgres, MongoDB, Neo4j, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised where agentic AI could support them and translated their requirements into the solution design.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector-search retrieval, general inference and summarization.
- Built the Python/FastAPI backend (APIs tested with Postman) and a React frontend, containerized with Docker and deployed to AWS with CDK; version control in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of RAG retrieval and generation with RAGAS and PytrecEval against chemist-provided ground truth, running as an evaluation pipeline on AWS (Lambda, SQS, DynamoDB) built with CDK for PrismaChat.
- Benchmarked multilingual embedding models for retrieval on over 1M chunks of internal chemical data and designed RAG pipelines (chunking strategies, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models on time-series process data with engineered lag and rolling-window features, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Order Inbox Agent | Python | Django | TypeScript | Postgres | LLMs

- AI agent that reads purchase orders from email bodies, PDFs and Excel files, extracts customer, items and quantities, runs a matching system that maps each line to a product catalog (exact, customer map, fuzzy, pgvector embeddings) and writes a sales order back to a mock ERP through a REST API.
- Django backend with Postgres and a TypeScript review screen where a sales rep confirms low-confidence matches; per-customer configuration and a labelled test set of orders run in CI to guard reliability.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA Fine-Tuning

- Fine-tuned an Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **AI Agents & LLMs:** AI agents (LangGraph, LangChain), LLM APIs (LiteLLM), RAG, embeddings, vector search, GraphRAG, MCP, prompt engineering, QLoRA fine-tuning, Hugging Face Transformers
- **Full-Stack & APIs:** Python, FastAPI, Python microservices, REST APIs (Postman), React (beginner)
- **Quality & Evaluation:** ground-truth evaluation, RAGAS, PytrecEval, Git, GitLab CI/CD
- **Data, Cloud & Production:** SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), Pandas, AWS (Lambda, SQS, DynamoDB, EC2, S3, CDK), Azure, Docker, Kubernetes
- **Ways of working:** AI-assisted development with Claude Code (custom skills, agentic coding workflows), Agile, clean code

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)